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Systemic Pathology for BSc Nursing (NS 2127)



Pathology of Gastrointestinal System

Oesophageal Varices

- Varices are tortuous dilated veins in the distal oesophagus and proximal stomach.
- Portal hypertension (Impair portal venous blood flow to liver)
>>> collateral bypass channels to divert portal blood flow through esophagus to superior vena cava >>> increased pressure in esophageal plexus >>> dilated tortuous vessels
- Varices usually produce no symptoms until they rupture, causing massive hematemesis

- Morphology
 - Tortuous dilated veins in submucosa of distal esophagus and proximal stomach
 - Protrusion of irregular mucosa into stomach lumen
- Clinical features
 - No symptoms until rupture
 - Rupture >>> massive hemorrhage into lumen

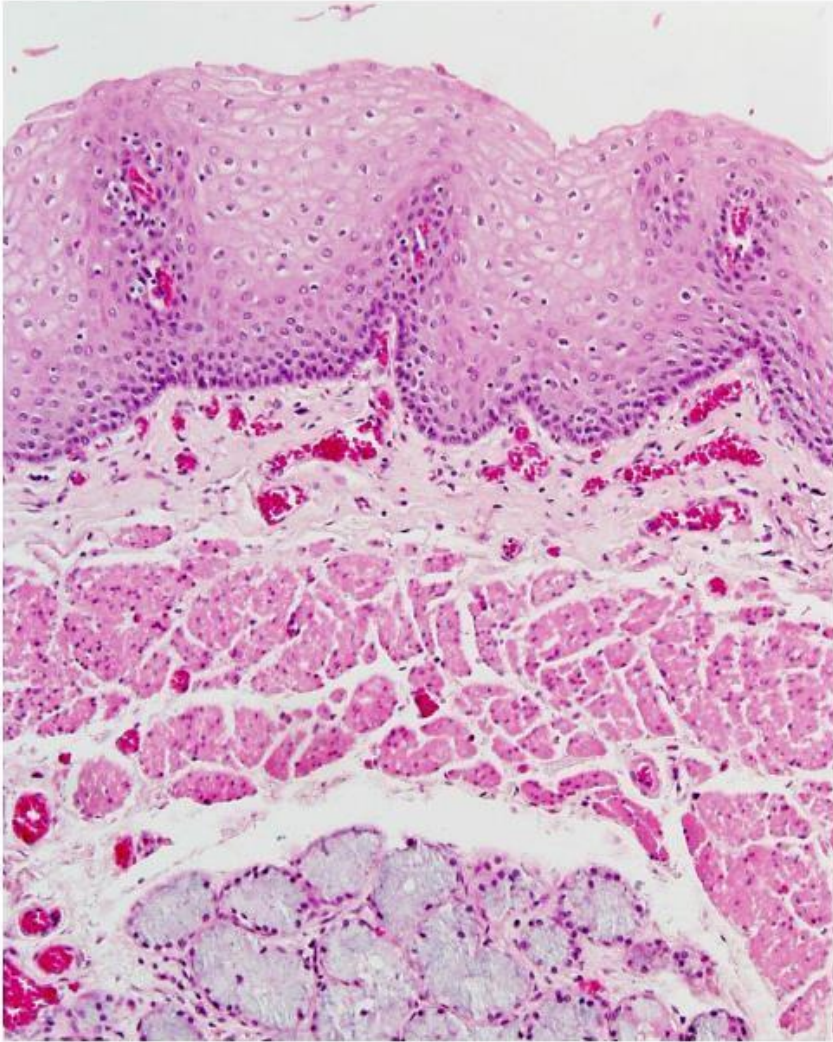
Oesophagitis

- Inflammation of esophageal mucosa
- Causes
 - Reflux of gastric content, prolong gastric intubation, uremia, ingestion of corrosive/ irritant substances, radiation and chemotherapy
- Mild to severe oesophagitis
- Clinical features
 - Heart burn accompany by regurgitation of sour brash
 - Rare – Chest pain

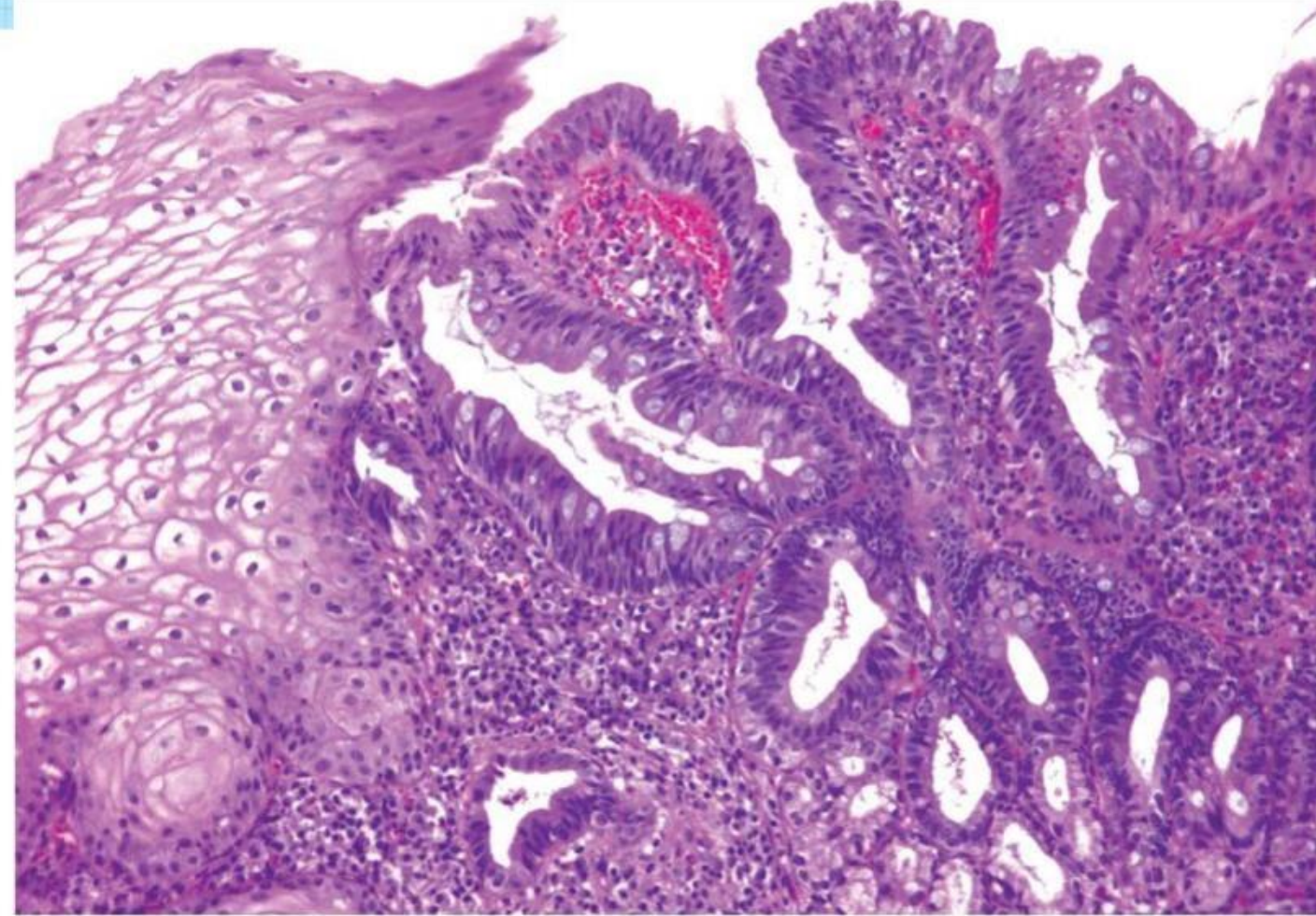
Berrett Oesophagus

- Replacement of the normal distal stratified squamous mucosa by metaplastic columnar epithelium containing goblet cells.
- Causes
 - Chronic gastric reflux
- Morphology
 - Salmon-pink velvety mucosa between normal smooth pale pink mucosa

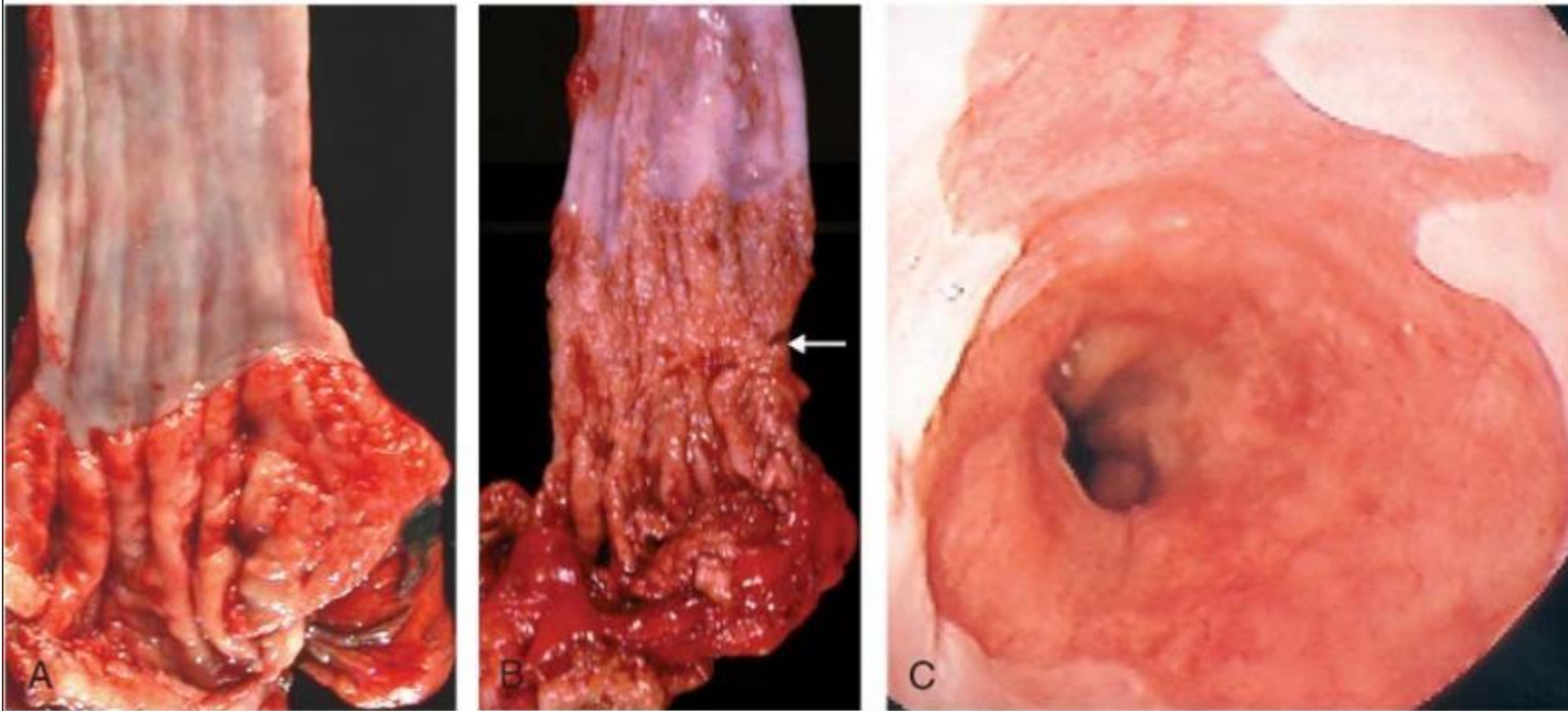
- Clinical features
 - Related to gastric reflux
- Complications
 - Ulcers and strictures



Esophagus: squamous mucosa



Barrett esophagus. Microscopic view showing squamous mucosa (left) and intestinal-type columnar epithelial cells in glandular mucosa (right).



Gross view of distal esophagus (*top*) and proximal stomach (*bottom*) showing (A) **normal gastroesophageal junction** and (B) the granular zone of Barrett esophagus (*arrow*). (C) Endoscopic view showing red velvety gastrointestinal-type mucosa extending from the gastroesophageal orifice. **Note paler squamous esophageal mucosa.**

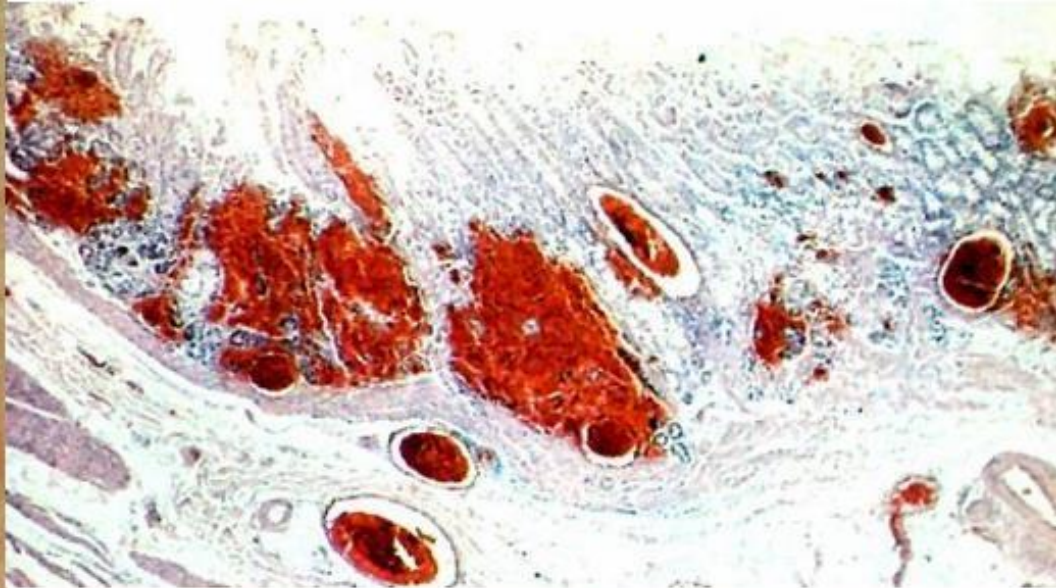
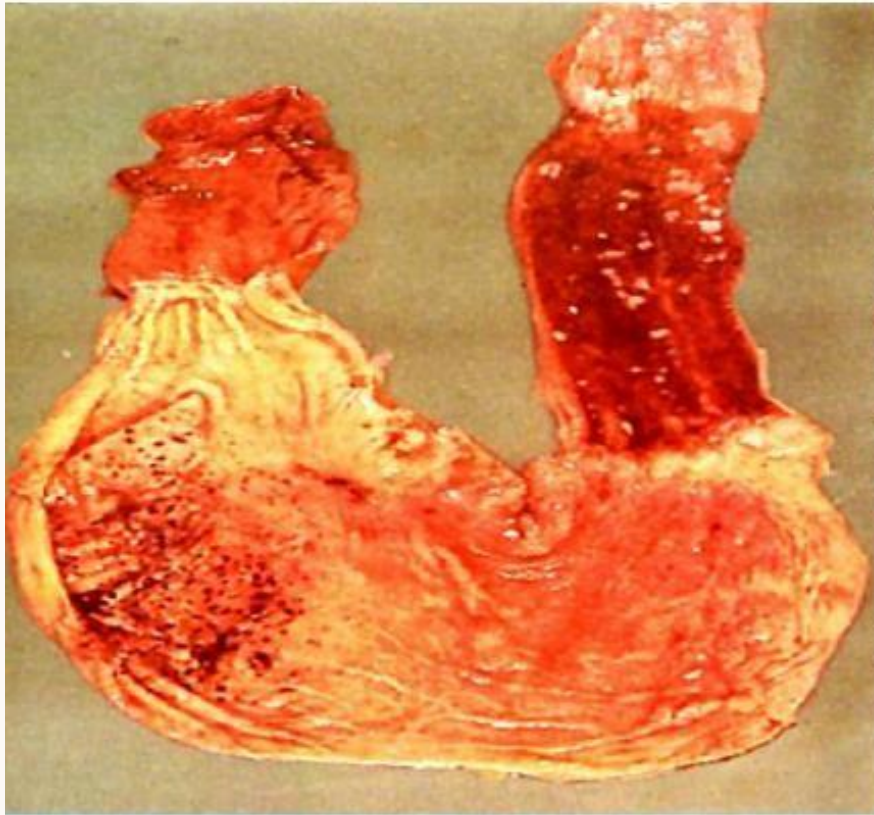
Acute Gastritis

- Acute inflammation, erosion, and hemorrhage of the gastric mucosa due to a breakdown of the mucosal barrier and acid-induced injury
- Transient in nature
- Causes
 - Aspirin or NSAID use, Alcohol use, Smoking, Post surgery, Ischemia, Stress, Uremia, Chemotherapy

- Morphology
 - Hemorrhage and focal erosions
 - Localized or diffuse, superficial to deep
- Clinical features
 - Asymptomatic, Epigastric pain with nausea and vomiting, hematemesis, melena, life threatening blood loss

(2) acute hemorrhagic gastritis

- 1) pathogens: medicines (aspirin)/alcoholism
- 2) lesion: usually in fundus and body



Chronic Gastritis

- Presence of chronic inflammatory changes in the stomach mucosa leading eventually to mucosal atrophy and epithelial metaplasia

- Causes

H Pylori chronic infection, Autoimmune, Chronic irritation,
Reflex of bilious duodenal secretion

- Morphology

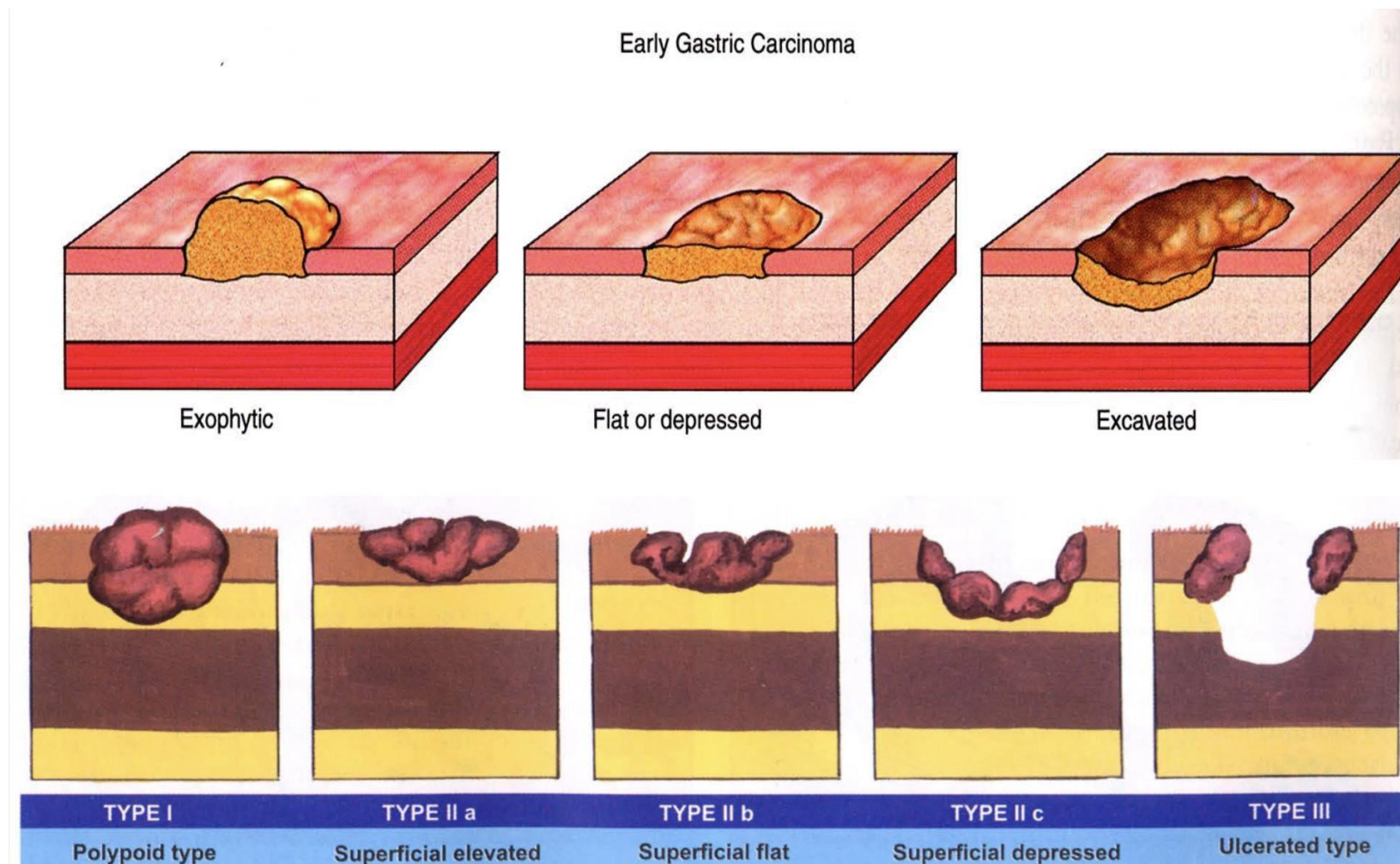
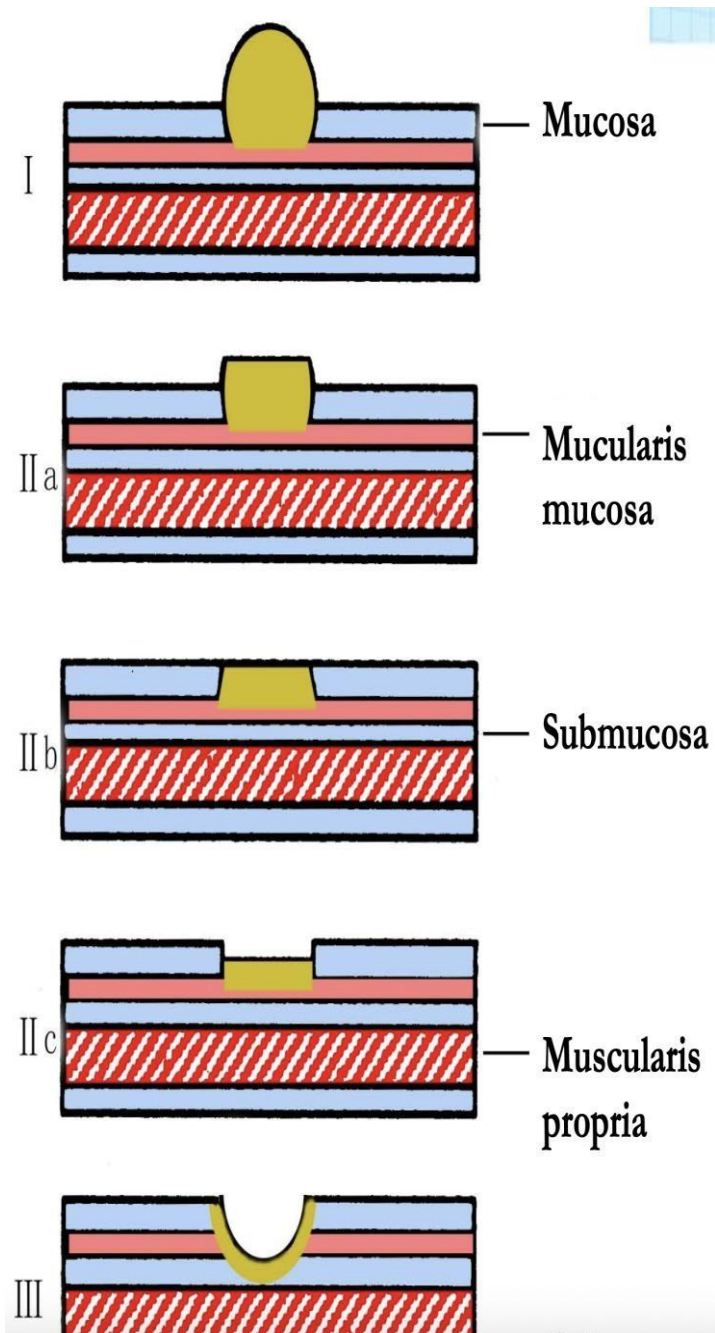
Lymphocytic and plasma cell infiltrates in lamina propria accompanied by neutrophilic inflammation, mucosal gland atrophy

- Clinical features
 - No symptoms, Upper abdominal discomfort, Nausea, Vomiting
- Complications
 - Increase risk of gastric carcinoma

Gastric Carcinoma

- Among the malignant tumors that occur in the stomach, carcinoma is overwhelmingly the most important and the most common (90% to 95%)
- Two morphologic types
 - Intestinal type; gastric mucous cells that have undergone intestinal metaplasia, better differentiated
 - Diffuse variant; arise de novo from native gastric mucous cells, ends to be poorly differentiated.

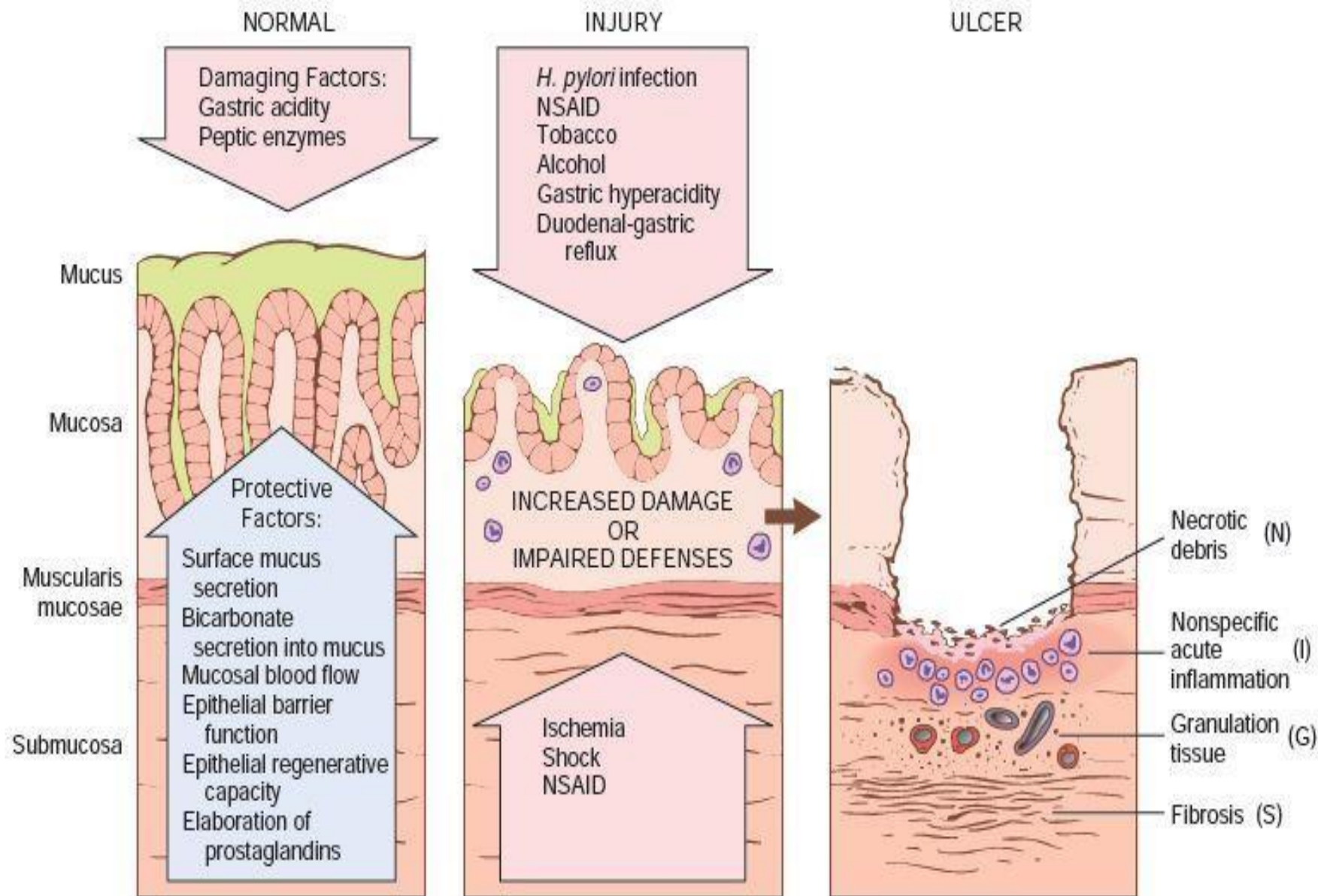
- Risk factors
 - Chronic gastritis with intestinal metaplasia, Infection with H pylori, Nitrites derived from food/ water, Diet (smoked food, pickled vegetables, excess salt), Decrease intake of fresh vegetables and fruits, Partial gastrectomy, Pernicious anemia
- Morphology
 - Favoured location is lesser curvature of antropyloric region, mass protrude into lumen/ flat or dressed lesion/ shallow or deep erosive crater
- Clinical features
 - Asymptomatic, Advanced carcinoma – abdominal discomfort and weight loss, may cause dysphagia



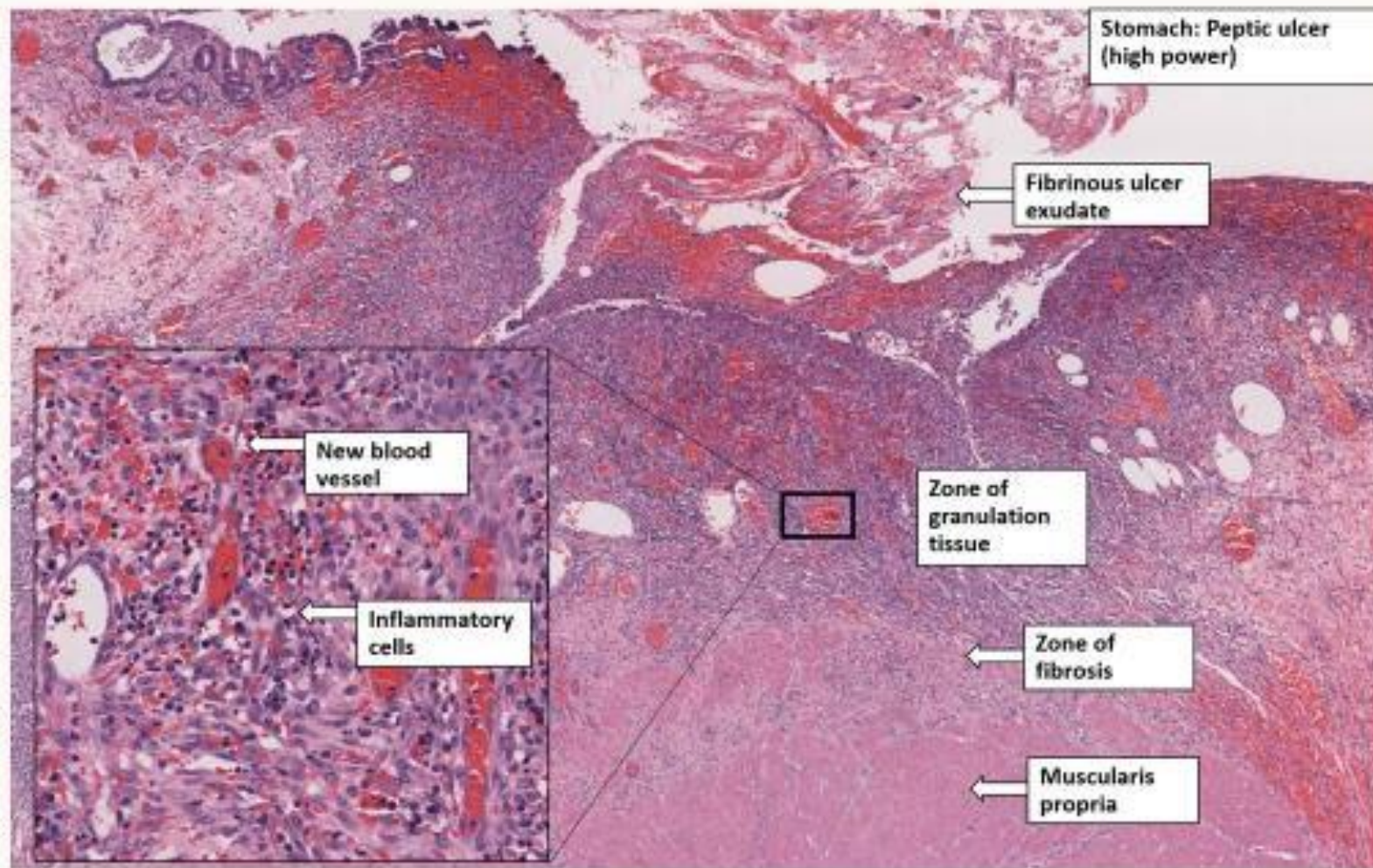
Peptic Ulcer Disease

- Chronic often solitary lesions occur in any portion of GIT exposed to aggressive action of acidic peptic ulcers
- Common Causes
H. pylori infection, NSAIDs, Medications
- Morphology
 - Defects in the mucosa that penetrate at least into the submucosa and often into muscularis propria or deeper
 - Round sharply punctured out craters 2-4 cm diameter

- Clinical features
 - Epigastric pain usually occurs within 15-30 minutes following a meal in patients with a gastric ulcer; on the other hand, the pain with a duodenal ulcer tends to occur 2-3 hours after a meal
- Complications
 - Bleeding, Perforation, Rare – obstruction and malignant transformations



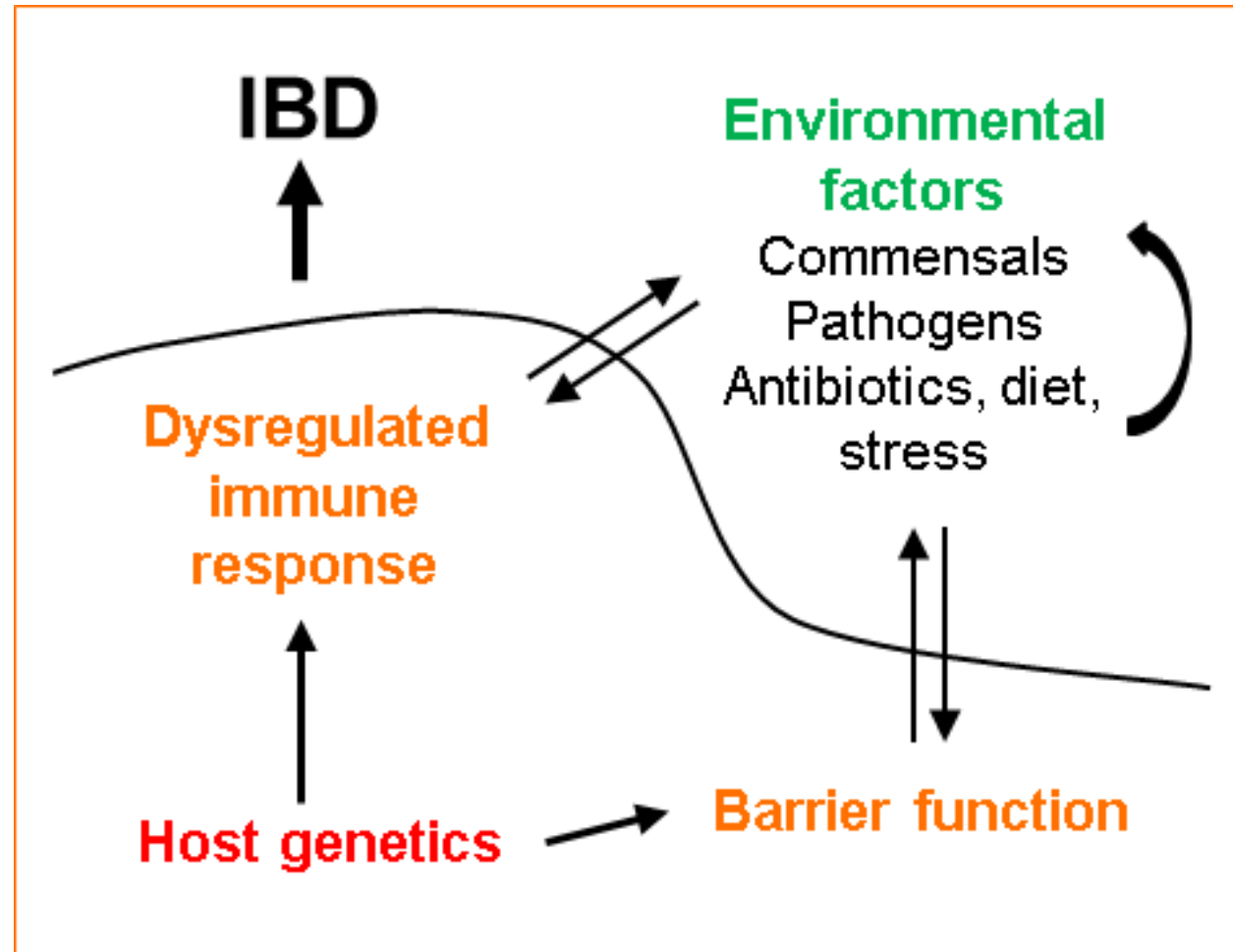
PEPTIC ULCER DISEASE



Inflammatory Bowel Disease

- Inflammatory bowel disease (IBD) is a chronic condition resulting from inappropriate mucosal immune activation
- The two disorders include are
 - Ulcerative colitis
 - Crohn disease

Pathogenesis



Clinical features

Feature	Crohn Disease	Ulcerative Colitis
Macroscopic		
Bowel region	Ileum $\pm$ colon	Colon only
Distribution	Skip lesions	Diffuse
Stricture	Yes	Rare
Wall appearance	Thick	Thin
Microscopic		
Inflammation	Transmural	Limited to mucosa
Pseudopolyps	Moderate	Marked
Ulcers	Deep, knife-like	Superficial, broad-based
Lymphoid reaction	Marked	Moderate
Fibrosis	Marked	Mild to none
Serositis	Marked	Mild to none
Granulomas	Yes (~35%)	No
Fistulae/sinuses	Yes	No
Clinical		
Perianal fistula	Yes (in colonic disease)	No
Fat/vitamin malabsorption	Yes	No
Malignant potential	With colonic involvement	Yes
Recurrence after surgery	Common	No
Toxic megacolon	No	Yes

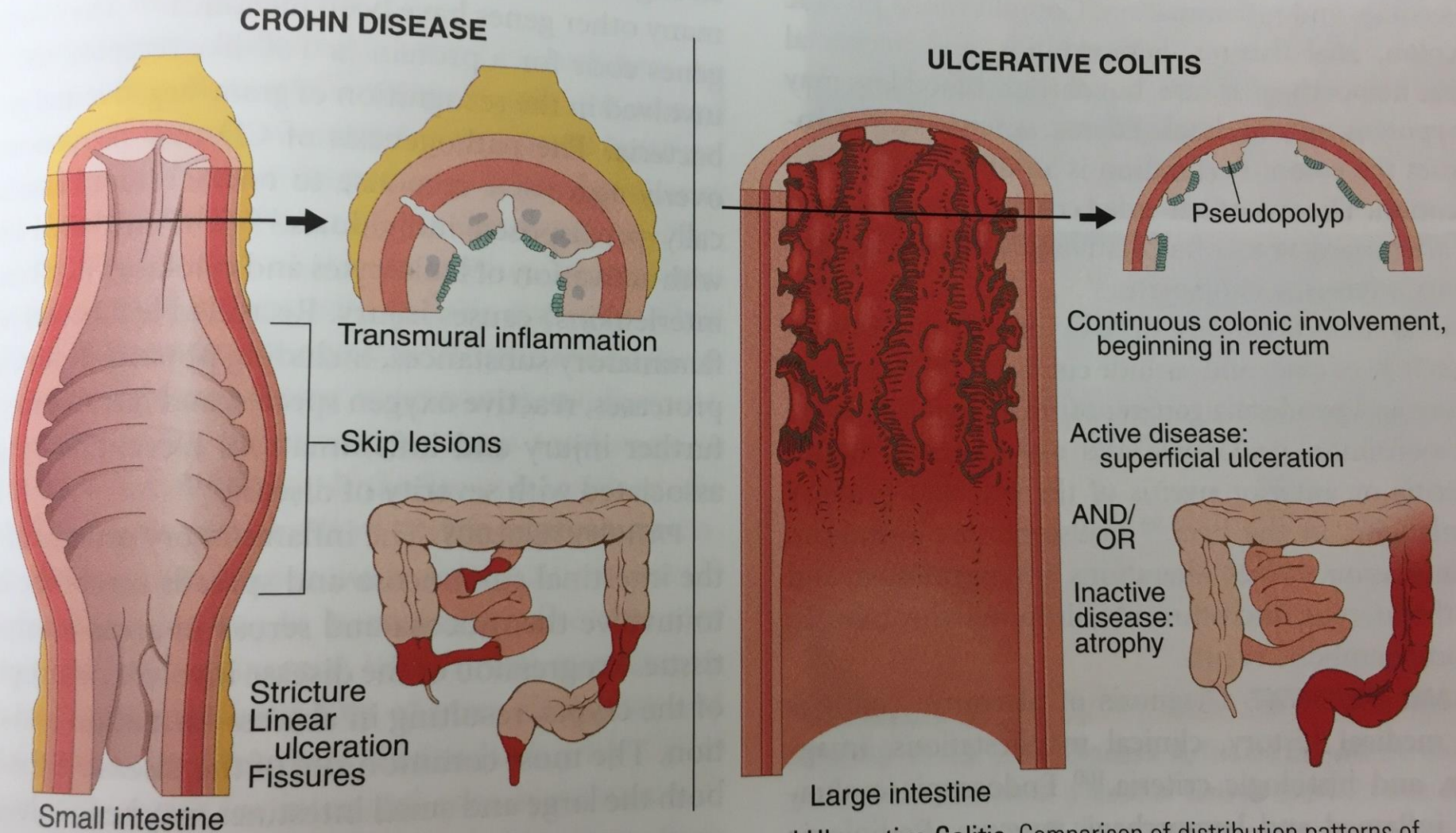


FIGURE 41-11 Distribution Patterns of Crohn Disease and Ulcerative Colitis. Comparison of distribution patterns of Crohn disease and ulcerative colitis as well as different conformations of ulcers and wall thickenings. (From Kumar V et al: *Robbins basic pathology*, ed 8, St Louis, 2008, Mosby.)

Symptoms

- Diarrhea - Sometimes mixed with blood, mucus and pus
- Cramping pains in the abdomen - Can be very severe and often occur before passing a stool
- Tiredness and fatigue
- Feeling generally unwell. Some people may feel feverish
- Loss of appetite and loss of weight
- Anemia

Complications

- Strictures, Fistula, Carcinoma, mostly with Ulcerative colitis

Colorectal Carcinoma

- Adenocarcinoma
- Different morphological patterns
 - Polypoid exophytic masses that extend along the wall
 - Annular encircling lesions (napkin-ring constriction and narrowing of lumen)
- Clinical features
 - Asymptomatic, Advanced - Fatigue, weakness, iron deficiency anemia, occult bleeding, change in bowel habits, crampy left lower quadrant discomfort

Hepatic Failure

- Most severe clinical consequence of liver disease
- Progressive damage to liver or sudden massive destruction of hepatic tissue >>>> Hepatic failure
- About 80-90% of hepatic function is lost >>> Hepatic failure
- Alterations causing liver failure
 - Acute liver failure with massive hepatic necrosis
 - **Chronic liver disease**
 - Hepatic dysfunction without overt necrosis

Clinical Features of HF in chronic liver disease

- Jaundice
- Hypoalbuminemia
- Hyperammonemia
- Palmer erythema
- Spider angiomas
- Hypogonadism and gynecomastia
- Weight loss
- Complications: Coagulopathy, Hepatic Encephalopathy, Hepatorenal syndrome



n engl j med 360;3 nejm.org january 15, 2009

Cirrhosis

- Diffuse process characterized by fibrosis and conversion of normal liver architecture to structurally abnormal modules
- Causes
 - Alcohol abuse
 - Chronic infections – Hepatitis B and C
 - Autoimmune hepatitis
 - Biliary diseases
 - Iron overload

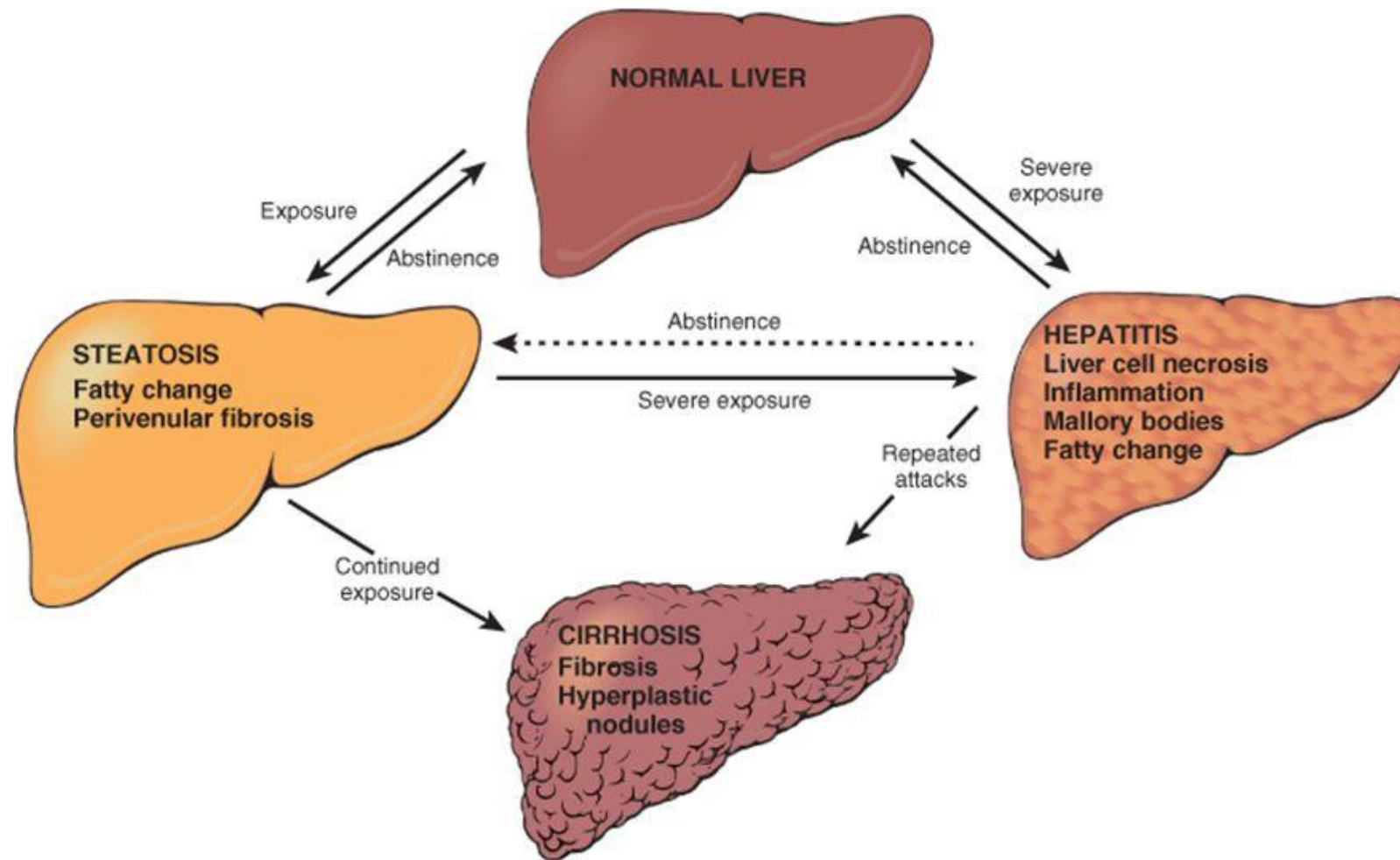
- Main characteristics
 - Bridging fibrous septa
 - Parenchymal nodules
 - Disruption of architecture of entire liver
- Clinical features
 - Clinically silent
 - Nonspecific symptoms – anorexia, weight loss, weakness
- Outcomes
 - Progressive liver failure
 - Portal hypertension and complications
 - Hepatocellular carcinoma

Portal Hypertension

- Increased resistance to portal blood flow
- Causes
 - **Cirrhosis**
 - Fatty changes etc.
- Consequences
 - Ascites
 - Portosystemic shunts
 - Congestive splenomegaly
 - Hepatic encephalopathy

Alcoholic Liver Disease

- Chronic alcohol consumption causes ALD
- Three forms
 - Hepatic steatosis (fatty liver) – 90-100% heavy drinkers
 - Alcoholic hepatitis – 10-35%
 - Cirrhosis – 8-20%



Alcoholic liver disease

The interrelationships among hepatic steatosis, hepatitis, and cirrhosis

Morphology of ALD

- Hepatic steatosis (fatty liver)
 - Large, soft, yellow and greasy liver, Lipid globules in hepatocytes
- Alcoholic hepatitis
 - Hepatocytes swelling and necrosis, Mallory bodies, Neutrophil infiltrates, Fibrosis
- Cirrhosis – 8-20%
 - Yellow, tan, fatty enlarged liver >>> Brown, shrunken, nonfatty
 - Micronodules

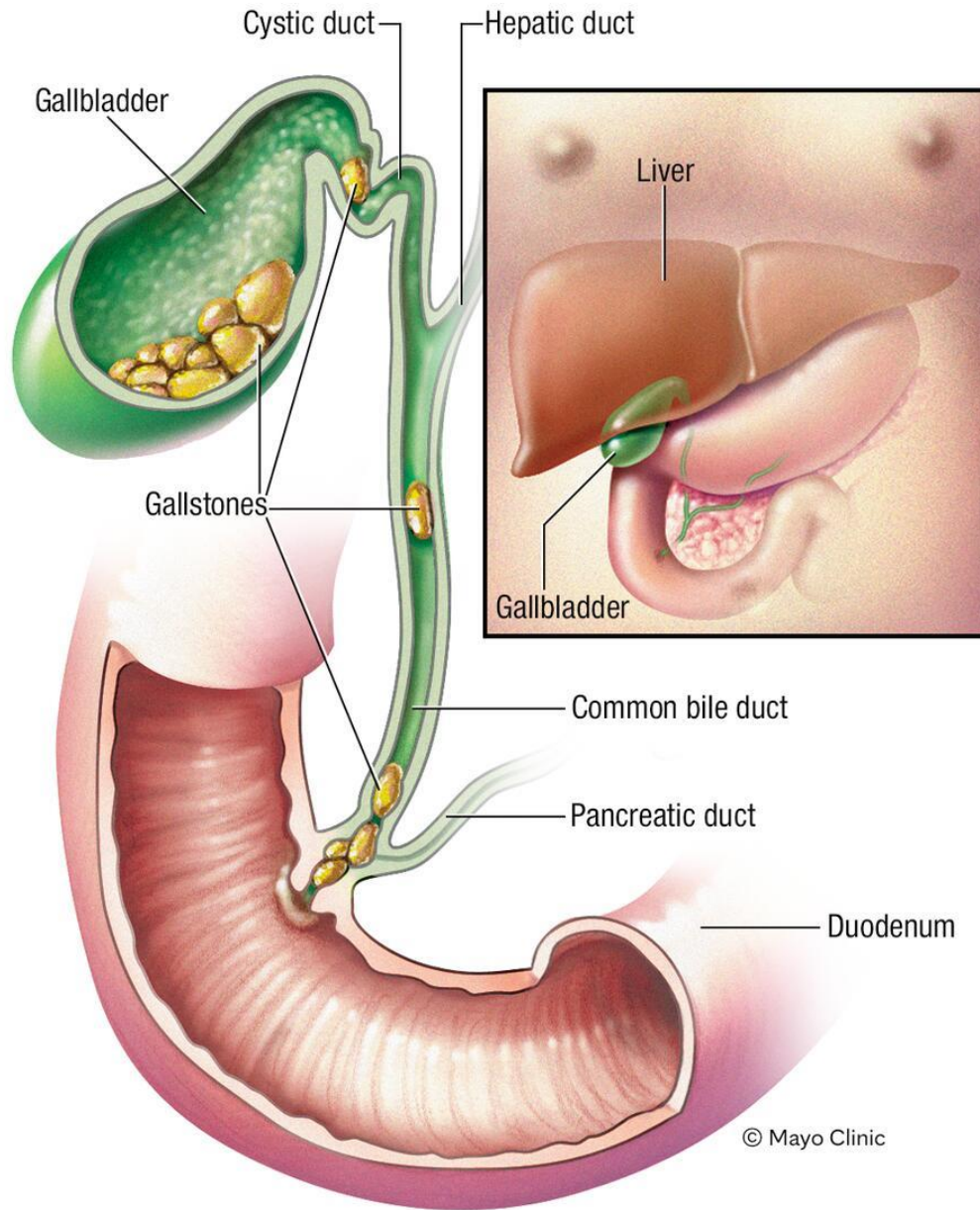
Hepatocellular Carcinoma

- 5.4% of all cancers
- Rarely present before the age of 60 years
- 90% of the cases are in patients with cirrhosis
- Factors involved; Infection with HBV or HCV, chronic alcoholism, aflatoxin exposure, Other factors – age, gender, hormones, alcohol, nutrition etc.

- Morphology
 - grossly unifocal (massive) tumor or multifocal tumor made of nodules of variable size or diffusely infiltrative cancer
- Clinical features
 - Often found in patients with cirrhosis
 - Rapid increase in liver size, sudden worsening of ascites, appearance of bloody ascites, fever, pain

Cholelithiasis

- Gallstones
- Two types; cholesterol, pigment stones
- Occurs as a result of
 - Supersaturation of bile with cholesterol
 - Formation of nucleation sites
 - Gallbladder stasis
 - Mucus hypersecretion
- Risk factors
 - Age, Gender, Ethnicity, Hereditary, Environmental, Acquired conditions



- Clinical signs

- Sudden and rapidly intensifying pain in the upper right portion of the abdomen/ center of the abdomen
- Back pain, pain in right shoulder
- Nausea or vomiting

- Complications

- Inflammation of the gallbladder
- Obstruction of common bile duct, pancreatic duct

- Treatment

- Surgical
- Nonsurgical – lithotripsy, Ursodiol